# Review of Magnetostatics and Faraday's Law

James T. Aberle, Ph.D.
*Associate Professor Emeritus*

School of ECEE

IRA A. FULTON SCHOOLS OF engineering
ARIZONA STATE UNIVERSITY

# Lecture Objective

- Provide a quick review of a topic covered in EEE 241: Magnetostatics and Faraday's law.

# Static Case

- Electric and magnetic fields become decoupled in the static case (i.e., there is no time variation):

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho_V \, dv$$

$$\oint_C \vec{E} \cdot d\vec{l} = 0$$

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{s}$$

# Magnetostatics

- ***Magnetostatics*** is the branch of electromagnetics dealing with the effects of electric charges in steady motion.

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{s} = I_{encl}$$

# Question

- Why is a static electric field called “irrotational” whereas a static magnetic field is called “solenoidal”?

# Electric Current

- Static (steady, not changing in time, “DC”) ***electrical current*** is the fundamental source of magnetostatic field.
- Steady electric current can occur as
  - volume current density ($A/m^2$) ⇐ **most general**
  - surface current density (A/m)
  - line current (A)

# Fundamental Postulates of Magnetostatics

$$\nabla \cdot \vec{B} = 0$$

Gauss's law for the magnetic field

$$\nabla \times \vec{H} = \vec{J}$$

Ampere's circuital law

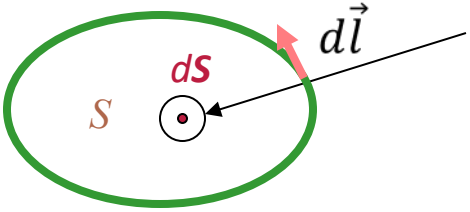
Helmholtz theorem tells us that we have now a complete specification of the magnetic field to within an additive constant which we assume is zero.

# Ampere's Circuital Law (1 of 2)

- Ampere's law states that "the circulation of the magnetic field is proportional to the total current through the surface bounding the path over which the circulation is computed."

$$\boxed{\oint_C \vec{H} \cdot d\vec{l} = I_{encl}}$$

# Ampere's Circuital Law (2 of 2)




By convention, $d\vec{s}$ is taken to be in the direction defined by the right-hand rule applied to $d\vec{l}$.

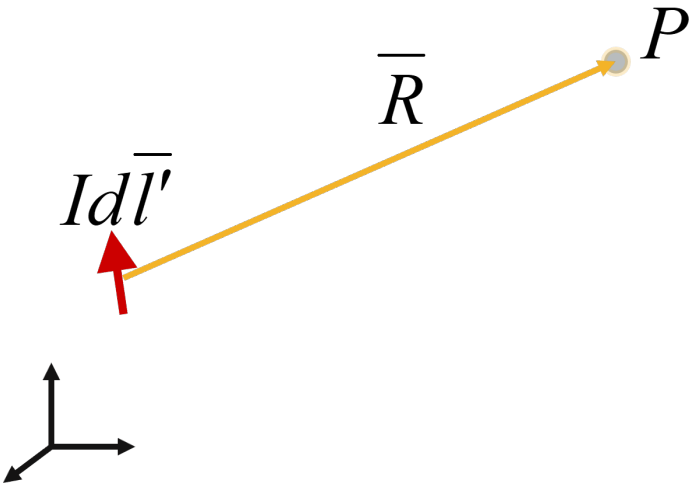
$$I_{encl} = \int_S \vec{J} \cdot d\vec{s}$$

Since volume current density is the most general, we can write $I_{encl}$ in this way.

# The Biot-Savart Law (1 of 2)

- The *Biot-Savart law* gives us the *magnetic field* arising from a given current distribution.
- The Biot-Savart law states the differential magnetic field generated by a steady current flowing through a differential length vector is:

$$d\vec{H} = \frac{1}{4\pi}\frac{d\vec{l} \times \hat{R}}{R^2}$$



# The Biot-Savart Law (2 of 2)

- The total magnetic flux at the point *P* due to the entire circuit *C* is given by

$$\vec{H} = \frac{I}{4\pi} \oint_C \frac{d\vec{l} \times \hat{R}}{R^2}$$

- For a surface and volume current distributions

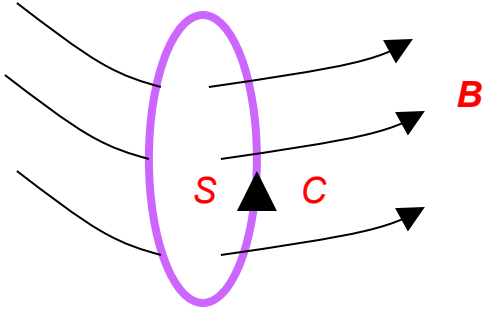
$$\vec{H} = \frac{1}{4\pi} \oint_C \frac{\vec{J}_s \times \hat{R}}{R^2} ds$$

$$\vec{H} = \frac{1}{4\pi} \oint_C \frac{\vec{J} \times \hat{R}}{R^2} dv$$

# Magnetic Flux (1 of 2)

- The magnetic flux crossing an open surface *S* is given by

$$\underbrace{\Phi}_{\text{Wb}} = \int_S \underbrace{\vec{B}}_{\text{Wb/m}^2} \cdot d\vec{s}$$



# Magnetic Flux (2 of 2)

- From the divergence theorem, we have

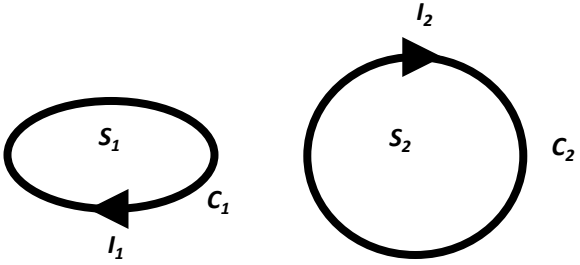
$$\nabla \cdot \vec{B} = 0 \Rightarrow \int_V \nabla \cdot \vec{B} dv = 0 \Rightarrow \oint_S \vec{B} \cdot d\vec{s} = 0$$

- Hence, *the net magnetic flux leaving any closed surface is zero*. This is another manifestation of the fact that there are no isolated magnetic charges.

# Flux Linkage

- Consider two magnetically coupled circuits.
- The magnetic flux produced $I_1$ linking the surface $S_2$ is given by

$$\Phi_{12} = \int_{S_2} \vec{B}_1 \cdot d\vec{s}_2$$




- If the circuit $C_2$ comprises $N_2$ turns and the circuit $C_1$ comprises $N_1$ turns, then the total flux linkage is given by

$$\Lambda_{12} = N_1 N_2 \Phi_{12} = N_1 N_2 \int_{S_2} \vec{B}_1 \cdot d\vec{s}_2$$

# Mutual Inductance

- The *mutual inductance* between two circuits is the magnetic flux linkage to one circuit per unit current in the other circuit:

$$L_{12} = \frac{\Lambda_{12}}{I_1} = \frac{N_1 N_2 \Phi_{12}}{I_1}$$

- Inductance is specified in units of henries.

# Self Inductance

- ***Self inductance*** is a special case of mutual inductance.
- The ***self inductance*** of a circuit is the ratio of the self magnetic flux linkage to the current producing it:

$$L_{11} = \frac{\Lambda_{11}}{I_1} = \frac{{N_1}^2 \Phi_{11}}{I_1}$$

- For an isolated circuit, we call the self-inductance, *inductance*, and evaluate it using

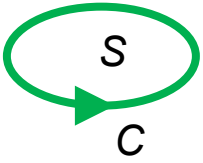
$$L = \frac{\Lambda}{I} = \frac{N^2 \Phi}{I}$$

# The Three Experimental Pillars of Electromagnetics

- Electric charges attract/repel each other as described by *Coulomb's law*.
- Current-carrying wires attract/repel each other as described by *Ampere's law of force*.
- Magnetic fields that change with time induce electromotive force as described by *Faraday's law*.

# Faraday's Law of Electromagnetic Induction

- "The electromotive force induced around a closed loop ***C*** is equal to the time rate of decrease of the magnetic flux linking the loop."
- Since the above must hold for any *S*, we have




$$V_{emf} = \oint_C \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

$$\boxed{\nabla \times \overline{E} = -\frac{\partial \overline{B}}{\partial t}}$$

# Faraday's Law and Lenz's Law

- Faraday's law states that a changing magnetic field induces an electric field.
- The induced electric field is *non-conservative*.
- Lenz's law states that "the sense of the emf induced by the time-varying magnetic flux is such that any current it produces tends to set up a magnetic field that opposes the <u>change</u> in the original magnetic field."
- Lenz's law is a consequence of conservation of energy and explains the minus sign in Faraday's law.

# Summary

- When there is no time variation, Maxwell's equations separate into uncoupled pairs of equations, one pair for the electric field and one pair for the magnetic field.
- The pair of equations for magnetostatics specify the curl and divergence of the magnetic field which are sufficient to completely specify the magnetic field.
- We can use Ampere's law to find the magnetic field for certain symmetric current distributions.
- We can use the Biot-Savart law to find expressions for the magnetic field due to any arbitrary current distribution.
- The inductance of a circuit is the ratio of the self magnetic flux linkage to the current producing it.
- Faraday's law describes how a time-varying magnetic field gives rise to a time-varying electric field.

Ira A. Fulton Schools of Engineering

Arizona State University